Yohei Ohata Curriculum Vitae

Center for Information and Neural Networks
National Institute of Information and Communications Technology
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Education

The University of Tokyo April 2022 – March 2024
M.S. in Human Factors Engineering
Department of Human and Engineered Environmental Studies
Cumulative GPA: 4.0/4.0
Thesis: Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models
1st in Class (1/48), Dean's Award, Best Department Thesis Award

Tokyo University of Science April 2018 – March 2022
B.E. in Electrical and Electronic Engineering
Department of Applied Electronics
Cumulative GPA: 3.5/4.0
Thesis: A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions

Academic Appointment

National Institute of Information and Communications Technology January 2026 – Present
Research Engineer
Center for Information and Neural Networks (CiNet)

The University of Tokyo July 2026 – Present
Cooperative Visiting Researcher
Graduate School of Information Science and Technology

Publications

Articles in Peer-Reviewed Journals

J1 **Ohata, Y.**, Yamashita, A., & Amano, K. (in preparation). Neural Substrates of Cognitive Control and Their Individual Variability. .

J2 **Ohata, Y.**, & Haruno, M. (in preparation). Convergent and Divergent Computational Principles of Context Processing in Transformer Attention and the Hippocampal Formation. .

J3 **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (submitted). Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models. *Frontiers in Psychology*.

Peer-Reviewed Conference Proceedings

C1 **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2024). Visualizing and Revealing the Difference of Learner's Thought Process and Question Solving Method. *In Proceedings of International Conference on Information and Communication Technology.*

C2 **Ohata, Y.**, Takao, K., Natori, T., & Aikawa, N (2021). A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions. *ICICE Technical Report.*

Conference Presentations

P1 **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2023). Analysis of How Learners Learn Using Graphic Questions. *The 42nd Annual Conference of Japanese Society for Educational Technology.*

P2 **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2022). Identifying Perceptual Factors that Influence Impression of a Teacher in a Classroom. *The 41st Annual Conference of Japanese Society for Educational Technology.*

Patents

T1 **Ohata, Y.**, Jin, Y. (2024). "Management Issues Estimation Device Using Generative Artificial Intelligence". JP-Patent.

T2 **Ohata, Y.**, Jin, Y. (2025). "Management Issues Estimation System Using Bayesian Network and Large Language Models". JP-Patent.

Research Experience

Research Engineer January 2026 – Present
National Institute of Information and Communications Technology
Laboratory: Computational Neuroscience & Decision Making Laboratory
Supervisor: Masahiko Haruno

- Investigating the degree to which attention mechanisms in Transformer architectures share functional and representational properties with hippocampal computation, and where the two diverge.
- Trained a single-layer Transformer with one attention layer on random walks over a community-structured graph, finding that hippocampus-like representations emerge spontaneously.
- Fitting a dynamical systems model to the Transformer's internal representations revealed that information retention time in attention heads increases with the number of communities, and inter-unit coupling strength grows as community transitions become more deterministic.

Cooperative Visiting Researcher August 2025 – Present
The University of Tokyo
Laboratory: Amano & Nakayama Lab
Supervisor: Ayumu Yamashita, Kaoru Amano

- Identified task-general and task-specific neural substrates of cognitive control at both between- and within-subject levels using public fMRI datasets (AX-CPT, Cued Task Switching, Sternberg, Stroop)
- Removed confounds via GLM-based nuisance regression and synchronized inter-subject timeseries using BrainSync (MATLAB)
- Decomposed the synchronized data into spatial, temporal, and subject modes via tensor decomposition, extracting multiple network components
- Identified task-general and task-specific networks based on inter-task correlations of spatial modes

- Identified networks governing inter-subject variability in cognitive control performance using subject modes and conditional randomization test
- Identified neural substrates underlying within-subject sustained attention states using temporal modes and GLMM

Research Assistant April 2022 – March 2024
The University of Tokyo
Laboratory: Inovative Learning Creation Lab
Supervisor: Satori Hachisuka, Shin'ichi Warisawa

- Designed experimental protocols for both adult and elementary school participants, including the development of a web-based UI and logging system using HTML/CSS/JS.
- Managed high-fidelity eye-tracking data collection and processing using Tobii Pro 3 and Tobii Pro Lab.
- Performed advanced statistical analysis using Linear Mixed-Effects Models (LMM) in R.
- Modeled the temporal dynamics of scan paths using Hidden Markov Models (HMM) implemented in Python.
- Conceptualized research goals, authored manuscripts.

Research Assistant May 2021 – December 2021
Tokyo University of Science
Laboratory: Signal Processing Lab
Supervisor: Naoyuki Aikawa

- Proposed a Composite Hilbert Transformer that integrates a pre-normalized notch filter to suppress noise at specific frequencies.
- Verified significant SNR improvements over conventional Band Pass Hilbert Transformer through extensive simulations in MATLAB.

Professional Experience

Research Engineer April 2024 – December 2025
Hitachi, Ltd., Digital Innovation R&D

- Developed concepts and business use cases
- Defined functional requirements and methodologies
- Authored patent applications
- Applied LLMs in service development
- Established evaluation criteria for LLM verification

Teaching Experience

Teaching Assistant January 2026 – Present
NICT CiNet, Computational Neuroscience & Decision Making Laboratory

- Delivered technical tutorials on Transformer internals and computational principles to 20+ lab members, covering QK/OV circuits and induction heads from a mechanistic interpretability perspective
- Introduced modern ML tooling (PyTorch, JAX, uv) and implementation practices to lab members
- Curated and presented 10+ landmark NeuroAI papers (past 5 years) to 20+ lab members, facilitating cross-disciplinary discussion between neuroscience and AI

Teaching Assistant April 2023 – June 2023
Sensing and Signal Processing, The University of Tokyo

- Assisted in building signal processing systems using LabVIEW

Teacher April 2019 – March 2022
Gakuho Seminar

- Taught Mathematics, Physics, Chemistry, and English to students in Years 10–18.
- Created and managed individualized study plans to support academic progress.
- Successfully guided over 30 students in passing their target school entrance examinations.

Honors and Awards

IEEE CAS JJC Best Student Award 2021

Languages

- High Level in English
- Native in Japanese

References

Masahiko Haruno, Ph.D.,
Professor
National Institute of Information and Communications Technology
Email: mharuno@nict.go.jp

Ayumu Yamashita, Ph.D.,
Associate Professor
Kobe University
Advanced Telecommunications Research Institute International
Email: ayumu@atr.jp

Robert Mok, Ph.D.,
Associate Professor
Osaka University
National Institute of Information and Communications Technology
Email: robmmok@gmail.com

Satori Hachisuka, Ph.D.,
Associate Professor
The University of Tokyo
Email: hachisuka@edu.k.u-tokyo